

Normalization, Relational Algebra & Indexing - Detailed Notes

DETAILED COMPREHENSIVE NOTES

Subject: Database Management Systems | Chapter: Chapter 2: Normalization, Relational Algebra & Indexing | Academic Year: 2026-2027

1. Functional Dependencies & Database Anomalies

Unnormalized relational designs suffer from modification anomalies: Insertion Anomalies (cannot insert record without unrelated data), Deletion Anomalies (loss of unintended data when deleting a record), and Update Anomalies (data redundancy leading to inconsistent state).

A functional dependency $X \rightarrow Y$ specifies that if two tuples agree on attributes X, they must agree on attributes Y.

2. Normal Forms (1NF through BCNF)

1NF (First Normal Form): All attribute values must be atomic (no multivalued or composite attributes, unique row identification).

2NF (Second Normal Form): In 1NF and no non-prime attribute is partially dependent on any candidate key (eliminates partial dependency).

3NF (Third Normal Form): In 2NF and no non-prime attribute is transitively dependent on candidate keys. Formally, for $X \rightarrow Y$, either X is a superkey or Y is a prime attribute.

BCNF (Boyce-Codd Normal Form): A stricter version of 3NF. For every non-trivial functional dependency $X \rightarrow Y$, X MUST be a superkey.

3. Relational Algebra & Database Indexing Structures

Fundamental Relational Algebra operators: Selection (σ), Projection (π), Cartesian Product ($\times$), Union ($\cup$), Set Difference ($-$), and Rename (ρ).

Database indexing accelerates query lookup. Primary indexes are built on ordered key fields; Clustering indexes on ordered non-key fields; Secondary indexes on unordered fields.

B+ Trees are the industry standard index structure: internal nodes store keys and child pointers, while all data pointers are stored exclusively in leaf nodes linked horizontally as a doubly linked list, optimizing range scans.